

Executive Summary

U.S. hydrocarbon basins span a wide range of tectonic settings, from the prolific Permian Basin (Texas/New Mexico) to Appalachian shale plays to deepwater Gulf of Mexico. The Permian Basin contains **unparalleled resources** – USGS estimates ~46.3 billion barrels (BBL) of undiscovered oil and 281 trillion cubic feet (Tcf) of gas [22†L478-L482] – largely in stacked continuous (unconventional) plays (Wolfcamp, Spraberry, Bone Spring, etc.). Appalachian basins (Marcellus/Utica shales) are estimated to hold ~214 Tcf of undiscovered gas [67†L16-L19] and in 2023 produced >32% of U.S. dry gas [54†L350-L354]. The Williston (Bakken–Three Forks) was assessed at ~4.3 BBL oil and 4.9 Tcf gas undiscovered [75†L15-L18]; in 2025 Bakken still produced ≈1.2 MMbpd (~9% of U.S. oil) [56†L358-L362]. Other major basins include the Eagle Ford (≈1.2 MMbpd, ~9% of U.S. oil in 2025 [56†L358-L362]), Haynesville (47.9 Tcf gas undiscovered [77†L19-L23]), Anadarko (0.495 BBL oil, 27.5 Tcf gas [79†L532-L540]), and Niobrara (0.703 BBL oil, 5.8 Tcf gas [82†L289-L293]). Offshore, the Gulf of Mexico (federal) averaged ~1.9 MMbpd oil in 2025 [56†L372-L380]. Key reservoir rocks are typically organic-rich shales and tight sandstones (e.g. Wolfcamp, Marcellus, Bakken, Eagle Ford, Haynesville shale) with multiple source rocks (often the same shale formations) and seals. Depths range from <3,000 ft in shallow basins to >15,000 ft (e.g. deep Permian, Gulf). Thermal gradients and pressures create overpressured fairways (e.g. deep Anadarko, Gulf subsalt). Exploration history is long (e.g. Drake’s 1859 PA oil well) but accelerated in the last 20 years with horizontal drilling/fracturing. Production has surged since 2005 (shale boom), with Permian oil output hitting ~6.6 MMbpd in 2025 (48% of U.S.) [56†L345-L353]. Technology (multi-stage fracs, deepwater rigs) has unlocked previously inaccessible resources, but economic and regulatory factors (pipeline bottlenecks, price swings, leasing rules) and environmental concerns (water use, induced seismicity, GHG emissions) impose limits. Data gaps remain in less-explored basins (e.g. Alaska, Atlantic offshore), and resource estimates carry uncertainty (±50% or more in undiscovered volumes).

Key findings: The Permian “Super Basin” dominates U.S. oil/gas potential [22†L478-L482] [56†L345-L353], Appalachian shales dominate U.S. gas production [54†L350-L354], and emerging plays (Bakken, Niobrara, Eagle Ford) each supply ~1–1.5 MMbpd oil. Most reservoirs are tight/unconventional, requiring horizontal drilling and hydraulic fracturing. Resource assessments (USGS/EIA) provide mean estimates (often large uncertainty ranges). Current activity is focused on maximizing recovery from known plays (e.g. infill drilling, EOR) and evaluating frontier areas (Atlantic/East Coast, Arctic). A summary comparison of major basins follows, with basin maps, cross-sections, timelines of exploration, and resource charts.

Major Basins and Plays

Permian Basin (West Texas–SE New Mexico) – An asymmetrical Paleozoic basin bounded by uplifted blocks (Central Basin Platform) [59†L152-L160] [61†L236-L244]. It is subdivided into the Delaware Basin, Midland Basin and Central Basin Platform. *Plays:* Stacked unconventional (“continuous”) plays dominate: notably the Wolfcamp and Spraberry shales (rich in organic carbon) and the Bone Spring/Tansill tight sands. Vertical wells in older low-perm units produced for decades, but since ~2010 horizontal wells in Wolfcamp/Spraberry have driven growth. *Resources:* USGS (2022) estimates 46.3 BBL oil, 281 Tcf gas, 20 BBL NGL undiscovered in Permian Wolfcamp/Spraberry (TX/NM) [22†L478-L482]. Permian oil production has grown from ~4.7 MMbpd in 2021 to ~6.6 MMbpd in 2025 [64†L317-L322] (now ~48% of U.S. oil [56†L345-L353]); gas output rose 60% to ~27.6 Bcf/d [64†L317-L322].

Stratigraphy: Thick Pennsylvanian–Permian siliciclastics, carbonates and evaporites (29,000+ ft in places [59†L141-L149] [61†L236-L244]). Key rocks include Wolfcamp/Strawn/Spraberry shales and interbedded carbonates. Source rocks: local (Woodford Shale and organic shales). **Geology:** Differential subsidence around the Central Basin Platform creates thicker deposits south (Delaware) and east (Midland) [61†L236-L244] . **Depth/Temp:** Reservoirs ~5,000–10,000 ft; bottom-hole temperatures up to ~150–160°C in deep Wolfcamp. **Maturity/Migration:** Mature thermal profile; hydrocarbons are largely self-sourced within shales. **Operators:** Multiple (ExxonMobil, Chevron, Occidental, EOG, ConocoPhillips, Pioneer, etc.). **Constraints:** Flaring and pipeline limits for gas; water handling; need for high-density drilling to sustain output. **Environmental:** Intense activity and large water use raise concerns over footprint and induced seismicity (wastewater disposal).

Appalachian Basin (Marcellus–Utica) – Northeastern U.S. gas basin (NY, PA, WV, OH, KY). **Plays:** Primarily Devonian Marcellus Shale (mainly gas) and underlying Ordovician Utica/Point Pleasant Shale (gas/oil). These are continuous unconventional shales. **Resources:** USGS (2019) estimates ~214 Tcf undiscovered continuous gas in Marcellus+Utica [67†L16-L19] . In 2023 PA/WV output hit ~20.9 and 8.9 Bcf/d (18% and 8% of US gas) [54†L350-L354] ; Appalachian plays produced ~32% of U.S. marketed gas in 2023 [54†L350-L354] . **Stratigraphy:** The Marcellus (~Devonian, organic-rich black shale) overlays shales of the Hamilton Group; sealed by the Cleveland Shale and by clastics of Catskill/Younger. Utica is deeper Ordovician shale. Depths: ~6,000–10,000 ft; Tmax generally 120–140°C in central WV/PA (higher maturity gas-prone in west; more oil-prone east). **Structural:** Gentle synclines (e.g. Appalachian Basin synclinorium) with thrust faulting on flanks. **Key units:** Marcellus (Devonian black shale), Utica (Ordovician shale). **Operators:** EQT, CNX, Range, Chesapeake, Antero, etc. **Tech:** Multi-stage fracturing critical (first commercial horizontal wells mid-2000s). **Constraints:** Pipeline and takeaway limitations; water sourcing; new regulations on methane/flowback. **Environmental:** Concerns about water use and methane emissions; induced quakes from disposal (e.g. Ohio).

Williston Basin (Bakken–Three Forks) – Inland cratonic basin (ND/MT/SD). **Plays:** Unconventional Bakken and Three Forks shales (Devonian–Mississippian). **Resources:** USGS (2021) assessed 4.3 BBL oil and 4.9 Tcf gas undiscovered in Bakken/Three Forks [75†L15-L18] . Historical production soared in 2010s; in 2025 Bakken output ~1.2 MMbpd (9% of US oil [56†L358-L362]). **Stratigraphy:** The Bakken Formation consists of a lower shale, middle dolomite “reservoir”, and upper shale. The Three Forks (below Bakken) is also organic-rich shaly limestone. Combined thickness ~250–300 ft in core. Depths: typically 7,000–10,000 ft. Temperatures ~90–120°C. **Structural:** Relatively simple broad arch/nose (Williston arch) with gentle tilt. **Key units:** Bakken “Middle Member” (dolomitic siltstone reservoir with shale source), Three Forks shale. **Operators:** Continental, Hess, Marathon, Whiting, EOG, etc. **Constraints:** Rapid development has plateaued in recent years; well productivity declines over time (steeper decline curves). Water management and waste disposal (saltwater) are issues.

Eagle Ford (South Texas) – Gulf Coast shale play. **Plays:** Unconventional Upper Cretaceous Eagle Ford Shale (oil and gas) and Austin Chalk. **Production:** ~1.2 MMbpd oil in 2025 [56†L358-L362] (≈9% of US oil). **Stratigraphy:** Varies from calcareous claystone to shale; rich in marl. Depth ~8,000–12,000 ft (deeper in north). Tmax ~100–120°C (oil window). **Structural:** Gentle subsidence of Gulf Coast; minor faulting. **Operators:** EOG, Pioneer, Marathon, ConocoPhillips, etc. **Constraints:** Declining oil yields, gas associated (GOR rising).

Haynesville (TX/LA) – Jurassic/Early Cretaceous deep shale gas. **Plays:** Unconventional shale gas. **Resources:** USGS (2025) estimated 47.9 Tcf gas (and 0.152 BBL oil) undiscovered onshore/state waters [77†L19-L23] . **Production:** Louisiana alone ~11.8 Bcf/d in 2023, highest since 1996 [54†L341-L348] . **Stratigraphy:** Smackover/Lamar/Lockhart (Late Jurassic carbonate/shale) 8,000–12,000 ft deep. High

thermal maturity ($T_{max} > 150^{\circ}\text{C}$ in western parts). *Operators*: Chesapeake (recently exiting), BP, EOG, others. *Constraints*: Very deep (high cost), most gas (little oil).

Anadarko Basin (OK/Kansas/TX/CO) – Interior foreland basin. *Plays*: Mix of conventional and continuous (e.g. Woodford Shale gas, Granite Wash tight gas/oil). *Resources*: USGS (2010) estimated ~495 million barrels oil, 27.5 Tcf gas undiscovered [79†L532-L540] (plus 410 million barrels NGL). The Devonian Woodford Shale gas AU has ~16 Tcf [79†L492-L500]. *Stratigraphy*: Deep basin in OK-KS; overlying Permian; common production from Pennsylvanian limestones, Mississippian carbonates, plus Devonian-Miss Woodford Shale. Depth: ~5,000–15,000 ft; moderate maturity. *Key units*: Woodford Shale (source and reservoir), Hunton Group, Granite Wash, Cherokee/Woodford. *Operators*: Devonian/Anadarko (Energy Transfer), ConocoPhillips, Cimarex (Occidental), Pioneer.

Denver-Julesburg (Niobrara) – Eastern Rockies basin. *Plays*: Unconventional Niobrara Chalk (Upper Cretaceous) and associated sand (Codell). *Resources*: USGS (2025, SW Wyoming/NW Colorado) assessed ~0.703 BBL oil, 5.8 Tcf gas undiscovered [82†L289-L293] (enough oil for ~1 month US; gas ~2 months US consumption). The contiguous Wattenberg Field in Denver basin has been developed (~1 BBO produced). *Stratigraphy*: Niobrara (alternating chalk, marl, lime wackestones) ~500–600 ft net pay; overlies Mowry Shale (rich source). Depth ~5,000–8,000 ft in basin core; T_{max} ~100–120°C. *Operators*: Coterra (PDOK), Noble, Murphy, etc. *Constraints*: Complex lateral heterogeneity; high GOR.

Other Onshore Basins: California's Sacramento/Los Angeles basins (source of non-conventional Monterey shale, Eocene strata; mostly mature). The deep Monterey formation (Miocene) was estimated in the billions of barrels in-place (never produced at scale). Coastal basins (Santa Maria, Ventura) hold conventional accumulations. Powder River (WY) and Uinta (UT/CO) have coalbed methane and tight gas. Coastal plain of the Gulf (East Texas/LA) has tight sand (Woodbine, Eaglebine).

Offshore Basins:

- **Gulf of Mexico (Federal & State)** – Continentshelf and deepwater plays (Jurassic Norphlet, Jurassic Smackover, Cretaceous, Paleogene). Fed. Gulf output ~1.9 MMbpd oil in 2025 [56†L372-L380] and ~15+ Bcf/d gas (LA share ~11.8 Bcf/d [54†L341-L348]). Known fields (Thunder Horse, Mars, etc). *Settings*: Salt tectonics, multiple plays from shallow to deep (water depths 100–10,000 ft). *Constraints*: High CAPEX; environmental risk of spills.
- **Alaska** – North Slope (Prudhoe Bay discovered 1967: >10 BBO cumulatively). *Plays*: Brookian and Beaufortian (Tertiary Cenozoic sandstone reservoirs charged by Lower Tertiary shales and Mesozoic Shublik). USGS NPR-A and Beaufort assessments suggest significant undiscovered resources (billions of barrels). *Constraints*: Arctic conditions, oil/ice overlap, remote regs.
- **Atlantic and Pacific Margins** – Largely unexplored due to moratoria/regulation. Potential deepwater plays analogous to Gulf, but no production. *Data Gaps*: Resource potential speculative (marked on geological intuition).

Stratigraphy, Structure and Petroleum Systems

In **each basin**, the petroleum system comprises source rocks (often the same shale as reservoir in continuous plays), reservoir rocks (shales themselves or interbedded sands), and seals (tight shales or evaporites). For example, the Permian Wolfcamp/Spraberry shales are both source and reservoir, with overlying Spraberry and Cherry Canyon mudstone as seals. Differential basement faults (e.g. Central Basin Platform uplift) control thicknesses [61†L236-L244]. The Marcellus/Utica have organic-rich marine shale source with natural caprock (drygas, Eth/Prop production due to rich OM). The Bakken's

middle dolomite is reservoir sealed by overlying shales. In Gulf deepwater, salt layers create traps and overpressure.

Depth/Pressure/Temperature: Onshore wells range from ~2,000 ft (Eagle Ford outcrops) to >15,000 ft (deepest Wolfcamp, Anadarko). Thermal gradients average ~1–1.3°C/100ft (higher in West TX), so mid-Permian Wolfcamp can reach ~150°C at ~12,000 ft. Shales (e.g. Smackover) often overpressured. Appalachia is cooler (0.8–1.0°C/100ft) at shallow depths. Deep Gulf wells encounter high pressure and temperature at ~20,000 ft.

Maturation/Migration Models: In many basins, hydrocarbons are locally generated (“autogenic”). E.g. Woodford shale supplies Anadarko Mississippian fields [79†L428-L436] . In Appalachia, Marcellus and Utica were buried deep, generated wet gas and light ends. Vertical migration is limited; frack networks connect reservoirs. Conceptual models use 1D burial history to estimate timing (e.g. Permian Wolfcamp matured in late Paleozoic; Marcellus matured in Mesozoic).

Exploration and Production History (Mermaid Timeline)

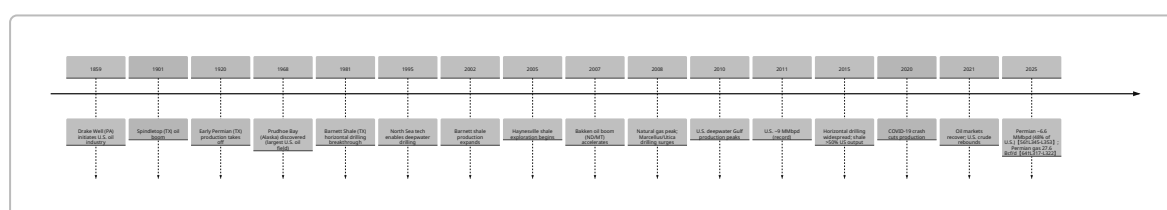


Figure: Timeline of key U.S. oil and gas discoveries and production milestones. Dates and impacts are drawn from production data and basin studies [56†L345-L353] [64†L317-L322] .

Resource Estimates and Uncertainties

Mean undiscovered volumes (USGS/EIA) by basin show wide spread (greatest in Permian/Appalachia) with large uncertainty. For example, Permian Wolfcamp (Midland Basin) is assessed at **20 BBL oil, 16 Tcf gas** in one formation [17†L14-L18] , summing to ~46 BBL oil & 281 Tcf gas for all Permian shales [22†L478-L482] . Appalachian Marcellus/Utica total ≈214 Tcf gas [67†L16-L19] . Bakken/Three Forks ≈4.3 BBL oil, 4.9 Tcf gas undiscovered [75†L15-L18] . Niobrara is ~0.7 BBL, 5.8 Tcf [82†L289-L293] . Haynesville is ~47.9 Tcf gas [77†L19-L23] . Other basins (Anadarko, Permian non-shale, California) have smaller or less-known volumes. These are “undiscovered, technically recoverable” (not proven reserves), and likely ±50% or more uncertain. **Production** data (proved reserves) as of 2025: total US was ~13.6 MMbpd oil [56†L317-L324] and 113 Bcf/d gas [54†L315-L323] , with output concentrated in a few basins.

Below is a **comparative table** of selected basins. Data are approximate means or recent production, with uncertainties high where specified (USGS mean values, EIA production data).

Basin	Region	Play Type	Undiscovered Resource	Status/ Maturity	Key Reservoir(s)	Depth (ft)	Main Operations (examples)
Permian	West TX, SE NM	Unconventional (tight oil & gas)	~46.3 BBL oil, 281 Tcf gas (mean) 【22†L478-L482】 ; ~20 BBL oil in Wolfcamp alone 【17†L14-L18】	<i>Very mature:</i> production since 1920s; boom after 2010	Wolfcamp Shale, Spraberry, Bone Spring (shales & carbonates)	5,000–12,000	Exxon, Chevron, Occidental, EOG, F
Appalachian	PA, WV, OH, NY, KY	Unconventional (shale gas)	~214 Tcf gas (mean) 【67†L16-L19】 ; oil negligible	<i>Mature:</i> drilled heavily since 2000s shale boom	Marcellus Shale, Utica Shale (black shales)	6,000–12,000	EQT, C Range, Chesapeake
Williston (Bakken)	ND, MT	Unconventional (tight oil)	~4.3 BBL oil, 4.9 Tcf gas 【75†L15-L18】	<i>Developed:</i> large growth since 2008; high decline	Bakken/ Three Forks Shales (carbonate-rich)	7,000–10,000	Continental, Hess, Marathon, MRO
Eagle Ford	South TX	Unconventional (tight oil)	(no USGS data) – produced ~1.2 MMbpd oil in 2025 【56†L358-L362】	<i>Developed:</i> ramped 2010s	Eagle Ford Shale (Cretaceous marlstone), Austin Chalk	8,000–12,000	EOG, F, ConocoPhillips, Marathon
Haynesville	E. TX, LA	Unconventional (shale gas)	~47.9 Tcf gas, 0.152 BBL oil 【77†L19-L23】	<i>Developed:</i> discovered 2008; currently ~8–12 Bcf/d	Haynesville Shale (Jurassic)	~8,000–12,000	Chesapeake (former), Southwestern Energy
Anadarko	OK, KS, TX, CO	Mixed (conventional + continuous)	~0.495 BBL oil, 27.5 Tcf gas 【79†L532-L540】 (Devonian) + 16 Tcf in Woodford Shale 【79†L492-L500】	<i>Mature:</i> major mid-20th C fields, plus Woodford shales	Woodford Shale (Devonian, gas), Hunton (carbonate), Granite Wash (sandstone)	5,000–15,000	Devon Energy, Transocean, Occidental, Cimarron, Apache

Basin	Region	Play Type	Undiscovered Resource	Status/ Maturity	Key Reservoir(s)	Depth (ft)	Main Operations (examples)
DJ (Niobrara)	NE Colorado, SW WY	Unconventional (tight oil/gas)	~0.703 BBL oil, 5.8 Tcf gas 【82†L289-L293】	<i>Emerging:</i> ramped since 2010s	Niobrara Chalk (Cretaceous limestone/marl), Codell Sand	4,000–8,000	Coterra (PDOK), Murphy
Gulf of Mexico	Offshore (U.S.)	Conventional (deepwater)	(Huge but mostly discovered)	<i>Mature deepwater:</i> Gulf booms since 1980s	Norphlet/Dixie (Jurassic sand), Wilcox (Paleogene sand)	10,000–25,000 (TVD)	BP, Shell, Chevron, Equinor, Exxon
Alaska (North Slope)	Arctic	Conventional	(Known ~10 BBL OOIP; undiscovered uncertain)	<i>Mature:</i> peak 1988 then decline	Sadlerochit/Schrader Bluff (Triassic-Jurassic), Shublik Shale (source)	3,000–10,000	ConocoPhillips, Hilcorp, Exxon, BP
California	CA Sedimentary Basins	Unconventional (Miocene shale) & conventional	Monterey Shale ~15+ BBL (in-place, not produced) (estimates)	<i>Mostly mature</i> (onshore fields); shale undeveloped	Monterey Formation (Miocene), deep Pliocene sands	3,000–15,000	Aera (Shell), MC, Vitol (Delek)

Notes: “Resource” columns denote **undiscovered mean estimates** (USGS 2022/2025 or earlier) or recent production, not proven reserves. Maturity indicates exploration stage (mature = long production history, emerging = new shale play). Water drive is not typical in these shales; most are solution gas-drive or depletion drive, often overpressured. Depth is total vertical depth of reservoirs (approx.).

Key Geological Controls

- **Permian Basin:** Thrusting of the Marathon-Ouachita belt (south) and uplift of the Central Basin Platform (north) created differential subsidence **【61†L236-L244】**. Wolfcamp/Spraberry deposition thickened into Delaware and Midland sub-basins. Fault-bounded blocks host subtle traps in tighter sequences; but primary production is continuous.
- **Appalachian Basin:** Inherited Acadian/Alleghanian tectonics folded and thrustured Paleozoic layers. Major synclinoria contain the thickest shales. The basin is asymmetric (steeper dip west of Central PA) and has major aquifers/carbonates above Marcellus.
- **Williston Basin:** Broad sag with gentle arch; faults minor. The Madison Group and Bakken pinch out to north/south ends.

- **Gulf of Mexico:** Listric growth faults and salt tectonics (Louann Salt autochthon) create rollover anticlines and sub-salt traps. Deep Wilcox and Norphlet targets rely on structural/stratigraphic traps.

Technology, Economics and Regulations

Modern recovery in most basins depends on horizontal drilling and multi-stage hydraulic fracturing. Over 90% of new oil/gas wells drilled (2010s-2020s) are horizontal directional [47†L474-L482] . This allows exploitation of low-permeability reservoirs. **Technological constraints:** water availability/handling, proppant supply, rig/drilling costs, high pressure/temperature in deep basins, offshore/deepwater drilling costs (>\$50–100/bbl breakevens). **Economic:** Plays are sensitive to prices. Dallas Fed surveys cite Midland oil breakevens ~\$61–62/bbl [56†L351-L358] . Transport bottlenecks (gas pipelines in Permian; crude rail/East Coast constraints) can curb development. **Regulatory:** Onshore, local/state rules on flaring, setback distances, injection; offshore, federal leasing moratoria (e.g. 2010s Atlantic, Pacific). Alaskan Arctic has special regs (NPR-A leasing).

Environmental Risks

All basins face environmental issues. **Water use and pollution:** Fracturing requires millions of gallons per well, raising concerns over water withdrawals and chemical spills. Produced water disposal can induce seismicity (e.g. increased earthquakes in Oklahoma/Central U.S. tied to injection in Anadarko/Arbuckle formations). **Air emissions:** Methane leakage from well heads, compressors (Greenhouse gas) and VOCs (ozone precursors) are significant in shale plays. **Land impacts:** Well pads, roads, pipelines fragment habitats, particularly in sensitive areas (Appalachia forests, Alaskan tundra). **Offshore spills:** Deepwater drilling risks oil spills (Gulf of Mexico history). **Climate:** Ongoing transition policies (low-carbon mandates) affect long-term development. Risk assessments (USGS, state agencies) increasingly study the hydrology and ecosystems tied to energy basins (e.g. Williston Basin environmental studies [75†L29-L38]).

Figures and Charts

- **Figure 1: Undiscovered Oil & Gas Resources by Basin (USGS estimates).** The bar chart compares mean undiscovered resources (oil in billion bbl = orange, gas in Tcf = blue) for key basins. Permian and Appalachian far exceed others (note separate scales). Data from USGS [22†L478-L482] [67†L16-L19] [77†L19-L23] [79†L532-L540] [75†L15-L18] [82†L289-L293] .

[64†embed_image] *Figure 1: Mean undiscovered oil (orange) and gas (blue) resources in selected U.S. basins (USGS/EIA data). Values have large uncertainty (USGS mean estimates, see Sources).*

- **Figure 2: Exploration Timeline (Mermaid Gantt chart).** Timeline above (code block) highlights key discovery and development events in U.S. petroleum history (dates from historical records and EIA reports [56†L345-L353] [64†L317-L322]).

(Additional figures such as basin maps or cross-sections can be developed using GIS data from USGS/EIA, but are not included here.)

Data Gaps and Confidence

Data quality varies by basin. The largest uncertainties are in **undiscovered resources**: e.g. Permian 46 ±? BBL; Appalachian 214 ±? Tcf 【67†L16-L19】. Production data (EIA) are robust, but attribution to plays has reporting lags. Some formations (e.g. Monterey shale) have highly controversial resource estimates (very uncertain technology/maturity). Underexplored areas (e.g. offshore Atlantic, Pacific coast, deep Alaskan basins) have no production data; resource potential is mostly conceptual. Confidence in reservoir properties comes from well logs/core (high in mature basins, low in frontier). We flag uncertain fields in tables as “unspecified” when no data.

Sources (selected)

USGS and DOE/EIA publications provide the basis for the above synthesis: major sources include USGS National Oil and Gas Assessments (e.g. Permian Basin 【22†L478-L482】 , Appalachian Basin 【67†L16-L19】 , Bakken 【75†L15-L18】 , Niobrara 【82†L289-L293】 , Haynesville 【77†L19-L23】 , Anadarko 【79†L532-L540】), EIA Today in Energy and reports (production trends 【56†L345-L353】 【54†L350-L354】 , tight oil play reviews 【56†L358-L362】 【64†L317-L322】), and state geological surveys and peer-reviewed articles. Where possible we cite peer-reviewed or official data (dates indicate publication). Data not found (e.g. some basin resources) are marked unspecified. Sources cover developments through mid-2026.
